



**FACULTY OF ENGINEERING, BUILT ENVIRONMENT, AND  
INFORMATION TECHNOLOGY (FOEBEIT)**

**BACHELOR OF COMPUTER SCIENCE - ODL**

**BCL1233 – SYSTEM ANALYSIS AND DESIGN**

**FINAL PROJECT**

**System Design and UML for a Hybrid Work Monitoring System**

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**INSTRUCTIONS TO CANDIDATE:**

1. This is an **INDIVIDUAL** assessment.
2. This assignment contributes **40%** of the total course marks.
3. This project builds upon the outputs developed in **Assignments 1, 2, and 3**.
4. All diagrams and design artefacts must be consistent with the requirements, process models, logic models, and data models developed in previous assignments.
5. Students may use AI tools to support idea generation and proofreading; however, all models, explanations, and design decisions must be original. Students must be able to explain and justify their work if requested.
6. Submit your completed assessment via **SPARK by 11:59 PM on 8 August 2026**.

## Learning Outcomes

|       |                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| CLO 1 | Analyse appropriate models, tools, and techniques used in object-oriented and structured systems analysis and design.                             |
| CLO 2 | Apply appropriate systems analysis and design processes in the development of a proposed information system solution.                             |
| CLO 3 | Design a system solution for non-routine problems in a real-world or simulated business environment using systems analysis and design techniques. |

## TASK DESCRIPTION

Based on the outputs developed in Assignments 1, 2, and 3, continue your role as a systems analyst to design a complete solution for the proposed **Hybrid Work Monitoring System**. The project should demonstrate the integration of requirements analysis, process modelling, logic modelling, data modelling, and system design. Students are expected to justify their design decisions and ensure consistency across all project artefacts.

## REPORT REQUIREMENTS

Students must prepare a structured report and respond to ALL sections below:

### Part A: User Interface Design (30 marks)

- Design low-fidelity wireframes for any **THREE (3)** of the Hybrid Work Monitoring System features below:
  - Submit Hybrid Work Request
  - Approve Hybrid Work Request
  - Employee Check-in/Check-out
  - Task Progress Update
  - Performance Dashboard
- Each wireframe must include:
  - Screen title
  - Navigation elements
  - Input fields
  - Action buttons
  - Error or confirmation messages
- Provide a brief explanation of each wireframe, including its purpose and key interface components.

### **Part B: UML Modelling (20 marks)**

1. Develop the following UML diagrams for **ONE (1)** selected system functions:
  - Sequence Diagram
  - Activity Diagram
2. All UML diagrams must be consistent with the requirements, process models, and logic models developed in previous assignments.
3. Provide a brief explanation for each UML diagram, describing its purpose and how it represents the selected system function.

### **Part C: Test Case Design (30 marks)**

1. Develop **FIVE (5)** test cases for **ONE (1)** selected system function.
2. Each test case should include:
  - Test Case ID
  - Test Objective
  - Input Data
  - Expected Result
  - Actual Result

Include both valid and invalid scenarios.

### **Part D: Traceability and Reflection (20 marks)**

1. Prepare a traceability matrix showing the relationships among the project artefacts.
2. Write a brief reflection that addresses the following:
  - a) Discuss the challenges encountered during the system analysis and design process.
  - b) Describe the approaches used to ensure consistency among the requirements, process models, data models, and UML diagrams.
  - c) Summarise the key lessons learned from developing the Hybrid Work Monitoring System.

## EVALUATION CRITERIA

| Assessment Component               | Description                                                                                                                                                                                                                                                                            | Mark Allocation |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>User Interface Design</b>       | Quality and appropriateness of the low-fidelity wireframes, including screen layout, navigation elements, input fields, action buttons, and messages. Designs demonstrate consistency with the system requirements and selected business processes.                                    | 30%             |
| <b>UML Modelling</b>               | Accuracy, completeness, and consistency of the UML diagrams, including the correct representation of relationships, interactions, and workflows. Diagrams demonstrate alignment with the requirements, process models, and logic models developed in previous assignments.             | 20%             |
| <b>Test Case Design</b>            | Quality, completeness, and relevance of the designed test cases, including clear test objectives, input data, expected results, and coverage of both valid and invalid scenarios. Test cases demonstrate alignment with the selected system function and its functional requirements.  | 30%             |
| <b>Traceability and Reflection</b> | Ability to demonstrate the relationships among the project artefacts through a traceability matrix. Reflection clearly discusses challenges encountered, approaches used to maintain consistency across models, and key lessons learned during the system analysis and design process. | 20%             |
| <b>TOTAL:</b>                      |                                                                                                                                                                                                                                                                                        | 100%            |

## SUBMISSION REQUIREMENTS

Students are required to submit the following upon completion of this assessment:

1. A complete report in **Word or PDF format**.
2. Structured answers covering all tasks and questions.
3. Clear headings and proper formatting for each section.
4. Optional diagrams or illustrations (if needed for explanation).
5. Ensure consistency with outputs developed in Assignments 1, 2, and 3.
6. Submission must be uploaded via **SPARK** on or before the given deadline.